

Computer vision in the food industry: Accurate, real-time, and automatic food recognition with pretrained MobileNetV2

Shayan Rokhva, Babak Teimourpour^{*,1}, Amir Hossein Soltani

Department of Information Technology Engineering, Faculty of Industrial and Systems Engineering, Tarbiat Modares University, Tehran, Iran

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ABSTRACT

In contemporary society, the application of artificial intelligence for automatic food recognition offers substantial potential for nutrition tracking, reducing food waste, and enhancing productivity in food production and consumption scenarios. Cutting-edge technologies such as Computer Vision and Deep Learning are highly beneficial, enabling machines to learn automatically, thereby facilitating automatic visual recognition. Despite some research in this field, the challenge of achieving accurate automatic food recognition quickly remains a significant research gap. Some models have been developed and implemented, but maintaining high performance swiftly, with low computational cost and low access to expensive hardware accelerators, is still an area that needs further exploration and improvement. This study employs the pre-trained MobileNetV2 model, which is cost-efficient and fast, for automatic food recognition on the public Food11 dataset, comprising 16643 images. It also utilizes various tools and techniques such as dataset understanding, transfer learning, data augmentation, regularization, dynamic learning rate, hyperparameter tuning, and consideration of images in different sizes to cut time and enhance the model's performance and robustness. These techniques aid in choosing appropriate metrics, achieving better performance, avoiding overfitting and accuracy fluctuations, speeding up the model, and increasing the generalization of findings, making the study and its results applicable to real-world applications. Despite MobileNetV2's simpler structure and fewer trainable parameters compared to some deep-dense models in the deep learning area, it quickly achieved approximately 92.97 % accuracy. This underscores the model's potential for implementation in real-world applications, which is the main intention of this study.

1. Introduction

AI-driven food recognition is crucial in the modern food industry, assisting in tasks like identifying restaurant foods, nutritional tracking, and classifying leftover or wasted food using image data (Lubura et al., 2022). AI visual recognition methods can be used not only for food recognition but also for agricultural applications like identifying and classifying plants, fruits, and vegetables, which are also linked to the food industry (Zhou & Chen, 2023). These applications can significantly boost efficiency and cut time in food production and consumption facilities (Chakraborty & Aithal, 2024; Lubura et al., 2022; Moumane et al., 2023). Empowered by computer vision and deep learning, advanced nations are adopting robotics in kitchens for efficient, minimal-waste operations. Their benefits are amplified by providing accuracy, speed, cost-effectiveness, and scalability for extensive real-world data (Chakraborty & Aithal, 2024; Fang et al., 2023;

Moumane et al., 2023). Moreover, the health benefits caused by mobile applications that regularly use AI for dietary purposes should not be neglected as they can lead to improved health for both present and future generations (Moumane & Idri, 2023).

Deep learning, a subset of AI, is superior at solving complex problems, especially with large datasets of the modern world. Convolutional Neural Networks (CNNs), blending deep learning and computer vision, are widely used for visual recognition. Their performance generally improves with greater datasets. CNNs use convolutional layers to automatically extract and aggregate image features, creating a rich feature map for image classification. They employ max pooling layers for feature selection and non-linear down-sampling, reducing computations. The final classifier determines the data prediction. Compared to CNNs, Artificial Neural Networks (ANNs), with their exponential increase in trainable parameters, require more computational costs, time, and storage, making them less suitable for image classification (Abiyev

* Corresponding author.

E-mail addresses: Shayan_rokhva@modares.ac.ir (S. Rokhva), B.teimourpour@modares.ac.ir (B. Teimourpour), Soltani.amirhossein@modares.ac.ir (A.H. Soltani).

¹ <https://orcid.org/0000-0002-9286-2286>.

& Adepoju, 2024; Lubura et al., 2022; Nath & Naskar, 2021).

The past decade has seen the development and application of various CNN models like AlexNet, VGG, GoogleNet, ResNet, EfficientNet, and MobileNet. Each model family's characteristics, size, depth, parameters, performance, and computational requirements differ. This shows that different models can suit different applications (Alzubaidi et al., 2021; Dhillon & Verma, 2020; Liu et al., 2024). Current studies usually focus on enhancing accuracy, reducing computational costs, and tailoring to specific applications (Alzubaidi et al., 2021; Karypidis et al., 2022; Silaparasetty, 2020).

Smart refrigerators, equipped with AI and IoT technologies, should accurately classify food items and proactively mitigate food waste by notifying owners about their freshness. Despite their current high cost, ongoing research indicates a promising future with more affordable and efficient models (Gao et al., 2019; Shweta, 2017; K. Wang et al., 2019). One such study focused on the classification of fruits such as oranges, apples, and bananas, determining their freshness or state of decay (Gao et al., 2019).

Food waste has drawn considerable public and policy attention. Alarming, 25–35 % of global food production remains uneaten depending on location, leading to resource wastage, greenhouse gas emissions, supply chain disruption, and financial burden on consumers (Ahmadzadeh et al., 2023; de Almeida Oroski & da Silva, 2023; Stenmarck et al., 2016). In this area, image processing techniques can be highly beneficial. Recognition, usually the initial step, is vital for identifying food or food waste types. Its accuracy greatly impacts subsequent processes and any inaccuracies could disrupt the entire operation (Farinella et al., 2020; Lubura et al., 2022; Moumane et al., 2023).

In this study (Kawano & Yanai, 2014), Deep-CNNs (DCNNs) were used to boost food recognition accuracy by integrating image features, color patches, and Fisher vectors. Another research (Kagaya et al., 2014) addressed the challenges of using classic CNNs and hyper-parameter tuning to enhance food recognition performance. A study utilized GoogleNet, a deep learning model with Inception modules, for food and non-food differentiation (Singla et al., 2016). This study (Zhu et al., 2020) developed a system using a video camera for recognizing stored fruits and vegetables. Based on Faster R-CNN, this model enhanced speed and accuracy over previous methods.

Transfer learning, a key deep learning tool, mitigates data scarcity, cuts computational needs, speeds up model training, and boosts performance. It includes feature extraction, adjusting only classifier parameters, fine-tuning, training deeper layers while early ones are fixed, and full fine-tuning, which trains all parameters and layers. Generally, there are significant differences between deep and dense models compared with lightweight and simpler models in terms of performance, time, computations, and hardware requirements. Some of these differences can be addressed by effective use of transfer learning (Dhillon & Verma, 2020; Farahani et al., 2021; Hosna et al., 2022; Iman et al., 2023).

The study (Siddiqi, 2019) used transfer learning for automated fruit image recognition with VGG16, a deep and dense model, achieving 99.27 % accuracy. They (Bansal et al., 2023) improved image classification using VGG19 and transfer learning, achieving 93.73 % accuracy on the Caltech-101 dataset with a Random Forest classifier, surpassing previous methods. They (Fakhrou et al., 2021) used DCNNs for smartphone-based food recognition, achieving 95.55 % accuracy with transfer learning. Deep ResNets (He et al., 2015) can generally be applied for accurate image classification. For example, to identify crop disease in the agriculture domain, having a significant impact on food security, ResNet34 achieved 99.40 % accuracy (Kumar et al., 2020). All (Bansal et al., 2023; Fakhrou et al., 2021; Kumar et al., 2020; Siddiqi, 2019) underscored the importance of transfer learning for cutting time and improving performance, specifically using dense models.

Lightweight models also need investigation. According to (Dong et al., 2020; Sahu et al., 2023), MobileNet's efficient architecture and computational performance make it suitable for real-time classification.

However, its computational efficiency might slightly compromise accuracy in more complex tasks. Nonetheless, this general statement does not always apply. For instance, using transfer learning and utilizing 108 images sized 256 * 256, MobileNetV2 was able to accurately classify soil images (Banoth & Murthy, 2024). In another study (Huang & Liao, 2022) LightEfficientNetV2, a lightweight CNN, with fewer parameters, highlighting its speed, achieved 98.33 % and 97.48 % accuracies on chest X-ray and CT images, respectively. This demonstrated that deep structures, requiring more computation and time, are not always the most accurate.

Quick and accurate image classification can be challenging due to shape, color, specific difficult patterns, and appearance similarities. But, CNNs and transfer learning have proven effective (Zhuang et al., 2020). VGG16, MobileNet2, and Resnet50 were tested on a public flower dataset, highlighting the difficulties in classifying similar flowers (Narvekar & Rao, 2020). This is something pivotal to consider as similar challenges regarding color, pattern or shape can occur in food categorization. Further explorations should address these challenges as solutions can positively affect the model's performance (Lubura et al., 2022; Mazlounian et al., 2020).

In this study (Lubura et al., 2022) a classic VGG-like CNN model was built from the ground up for food recognition, achieving 98.8 % accuracy. With image segmentation, Serbia's food waste was also found to be 21.3 %. In another study, using other image-processing techniques, with transfer learning data augmentation, they classified and segmented food waste with 83.5 % and 98.5 % accuracy and pixel accuracy, respectively (Mazlounian et al., 2020). Various models were used for waste detection, segmentation, and volume estimation, also considering organic waste such as food waste. Simpler and lightweight models were also found to be beneficial (Geetha et al., 2022).

Despite advancements in automatic food recognition, the challenge remains to achieve fast, accurate results with minimal computational cost and less dependency on expensive hardware accelerators, particularly for efficient deployment in real-world embedded and portable devices (Ahmadzadeh et al., 2023; Dong et al., 2020). This research aims to enhance accurate and real-time food identification using an optimized lightweight model like MobileNetV2. A wide range of deep learning techniques are used to enhance the model's robustness and performance and reduce time, in turn filling a gap and contributing to real-world applications.

This research's key contributions include:

1. Efficient application of MobileNetV2, a CNN model known for low computational demands and faster processing.
2. Utilizing the public Food11 dataset from Kaggle, with 16643 images across 11 food categories, highlighting noticeable inter-category similarities and intra-category diversities.
3. Implementing transfer learning in full fine-tuning mode to enhance performance, accelerate the model, save time, and address data scarcity.
4. Application of data augmentation, regularization, and hyper-parameter tuning to prevent overfitting, stabilize accuracy, and improve real-world applicability.
5. Consideration of different food image sizes to balance metric satisfaction and speed.
6. Detailed discussion offering valuable insights into the findings and suggestions for further improvements.

The rest of the study is organized as follows: Section 2 explains the methodology, Section 3 provides results and analysis, Section 4 discusses findings deeply, and Section 5 concludes the paper.

2. Materials and methods

2.1. Schematic diagram of the research workflow

Fig. 1 provides a visual summary of our research methodology, highlighting the key stages. The model is trained using the training dataset, leveraging transfer learning to utilize existing knowledge. The validation dataset is employed for hyperparameter tuning to create the optimal model. This model is then tested on the evaluation dataset, representing unseen data to demonstrate the model's real-world performance. Importantly, the model remains uninformed about the test/evaluation data until the inference mode.

2.2. Hardware Configuration

This research utilized Google Colab's T4 GPU, which offers 16 GB of memory and is an affordable option for many AI-driven applications.

2.3. Data acquisition, distribution & comprehension

Dataset comprehension is influential for effective research and analysis (Alvi, 2023; X. Wang & Zhu, 2023). The Food11 dataset, sourced from Kaggle (Food-11 image dataset, n.d.), had originally been divided into training, validation, and evaluation subsets, preventing data leakage (Brookshire et al., 2024). These subsets contain 9866, 3430, and 3347 images across 11 categories.

Table 1 breaks down the Food 11 dataset. It shows the number of images per food category and subset, such as 994 and 362 "Bread" images in training and validation, respectively. It also presents the distribution of each food category across subsets (e.g., "Dairy Product" is 59.6 % in training, 20 % in validation, and 20.5 % in evaluation) and different food categories within a specific subset (e.g., "Bread" is about 10 % of all training data while "Dairy-Products" are 4.3 %).

Table 1 indicates a 60 % allocation of all food category samples to the training subset, with the remainder divided almost equally between validation and evaluation. However, an imbalanced distribution is observed within each subset and the dataset. For example, "Rice" accounts for less than 3 % of images in all subsets, while "Meat" and "Dessert" make up about 13 % and 15 % respectively. This imbalance, particularly in the training subset, could affect the model's learning for certain categories. Therefore, further exploration is required via a

Table 1

Data distribution of the Food 11 dataset.

	Training	Validation	Evaluation	Total / Sum
Bread	994	362	368	1724
(in each class)	(57.6%)	(21%)	(21.4%)	(100%)
(in each subset)	(10%)	(10.5%)	(11%)	(10.4%)
Dairy Product	429	144	148	721
(in each class)	(59.5%)	(20%)	(20.5%)	(100%)
(in each subset)	(4.3%)	(4.2%)	(4.4%)	(4.3%)
Dessert	1500	500	500	2500
(in each class)	(60%)	(20%)	(20%)	(100%)
(in each subset)	(15.2%)	(14.6%)	(14.9%)	(15%)
Egg	986	327	335	1648
(in each class)	(59.8%)	(19.9%)	(20.3%)	(100%)
(in each subset)	(10%)	(9.5%)	(10%)	(9.9%)
Fried Food	848	326	287	1461
(in each class)	(58%)	(22.3%)	(19.7%)	(100%)
(in each subset)	(8.6%)	(9.5%)	(8.6%)	(8.8%)
Meat	1325	449	432	2206
(in each class)	(60.1%)	(20.3%)	(19.6%)	(100%)
(in each subset)	(13.4%)	(13.1%)	(12.9%)	(13.2%)
Noodles-Pasta	440	147	147	734
(in each class)	(60%)	(20%)	(20%)	(100%)
(in each subset)	(4.6%)	(4.3%)	(4.4%)	(4.4%)
Rice	280	96	96	472
(in each class)	(59.4%)	(20.3%)	(20.3%)	(100%)
(in each subset)	(2.8%)	(2.8%)	(2.9%)	(2.8%)
Seafood	855	347	303	1505
(in each class)	(56.8%)	(23.1%)	(20.1%)	(100%)
(in each subset)	(8.7%)	(10.1%)	(9.1%)	(9%)
Soup	1500	500	500	2500
(in each class)	(60%)	(20%)	(20%)	(100%)
(in each subset)	(15.2%)	(14.6%)	(14.9%)	(15%)
Vegetable-fruit	709	232	231	1172
(in each class)	(60.5%)	(19.8%)	(19.7%)	(100%)
(in each subset)	(7.2%)	(6.8%)	(6.9%)	(7.1%)
Total / Sum	9866	3430	3347	16643
(in each class)	(59.3%)	(20.6%)	(20.1%)	(100%)
(in each subset)	(100%)	(100%)	(100%)	(~100%)

confusion matrix analysis.

2.4. Data transformation, augmentation, and visualization

This study used two types of data transformations: Main and Augmenting. The Main transformation, including Resizing the image into its target image size (which are going to be 32, 64, 128, and 256 in this

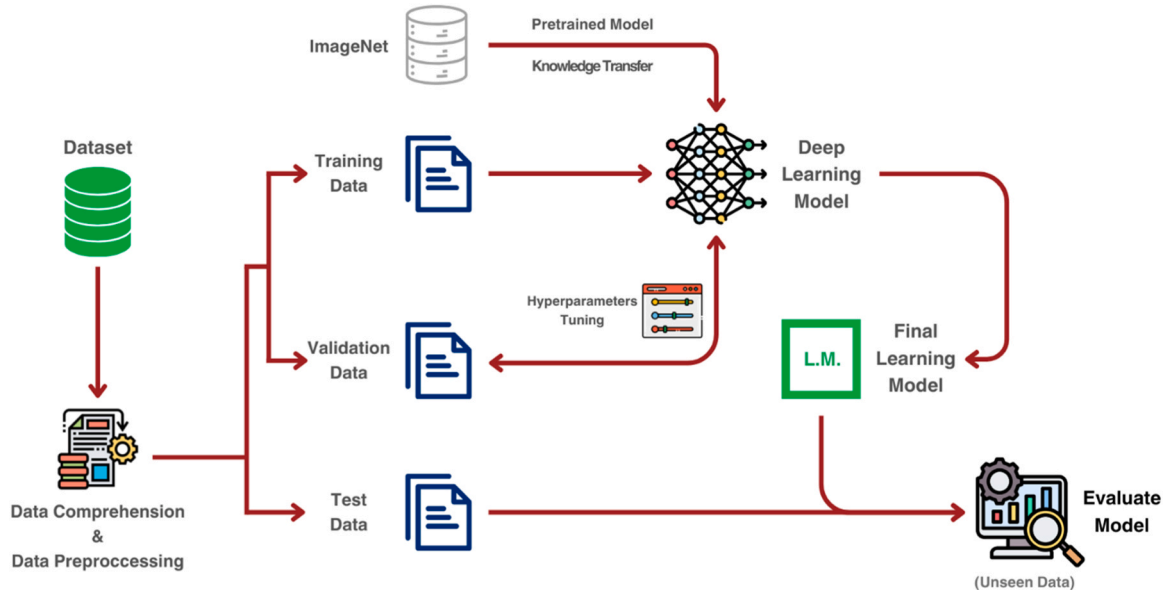


Fig. 1. General visual summary of the research methodology.

study), Tensoring (converting images into numerical tensors for interpretation), and Normalization, was applied to validation and evaluation subsets. The Augmenting transformation, which includes all main transformations with additional techniques like Random Rotation up to 30 degrees, Random Horizontal Flipping of 50 % of images, slight Color Jittering, and Random Erasing of 50 % of images, was used only during training to enhance data diversity and not quantity.

The main purpose of employed data augmentation was to improve model performance, and robustness, and to reduce overfitting by making the AI model train on more varied data. Moreover, we utilized Transfer Learning and existing knowledge to boost performance, and also Regularization to avoid overfitting so as to achieve enhanced generalization. Therefore, very strong and extensive data augmentation may not be required.

During Normalization, transfer learning allowed the use of ImageNet's mean and Standard deviation (STD). The technique did not exactly normalize the training set's mean and STD to 0 and 1 but approximated them closely, suitable for Food11 dataset images, which are in red, green, and blue channels (RGB), and similar to ImageNet data (Huh et al., 2016).

The study also explored how image resolution (32, 64, 128, 256) affects model speed and performance. As per Fig. 2, larger images enhanced clarity, but the clarity difference between two consecutive resolutions reduced as image size grew. High-resolution images (512 and 1024) were avoided due to their adverse impacts on model speed (Thambawita et al., 2021), particularly for real-time and non-sensitive applications such as food recognition.

2.5. Model

MobileNetV2 was employed due to its efficiency and speed caused by a combination of depth and point-wise convolution, linear bottlenecks, and shortcut connections. In many applications, it could outperform MobileNetV1 while having fewer parameters and computations. Furthermore, it is generally more popular than other MobileNet models (Ai et al., 2021; Sandler et al., 2018; Wu et al., 2023).

Fig. 3 shows MobileNetV2 has the fewest parameters (~3.5 million) among the mentioned ImageNet classification models, making it ideal for our study. Our customized version for the 11-class problem had 2.238 million parameters due to 11-class classifier modification, which made it even more lightweight.

2.6. Transfer learning

Transfer learning was used to enhance model performance and speed, conserve computational resources, and even address probable data scarcity. It is most effective when source and target datasets have similar features, and with enough data in the understudying dataset, full fine-tuning usually yields superior performance as it is more customized (Hosna et al., 2022; Zhuang et al., 2020). Therefore, we used a pre-trained MobileNetV2 model, customized it for an 11-class task, and benefited from full fine-tuning across 30 epochs. Given the similarities between Food11 and ImageNet, and that both datasets are RGB, effective transfer learning was expected.

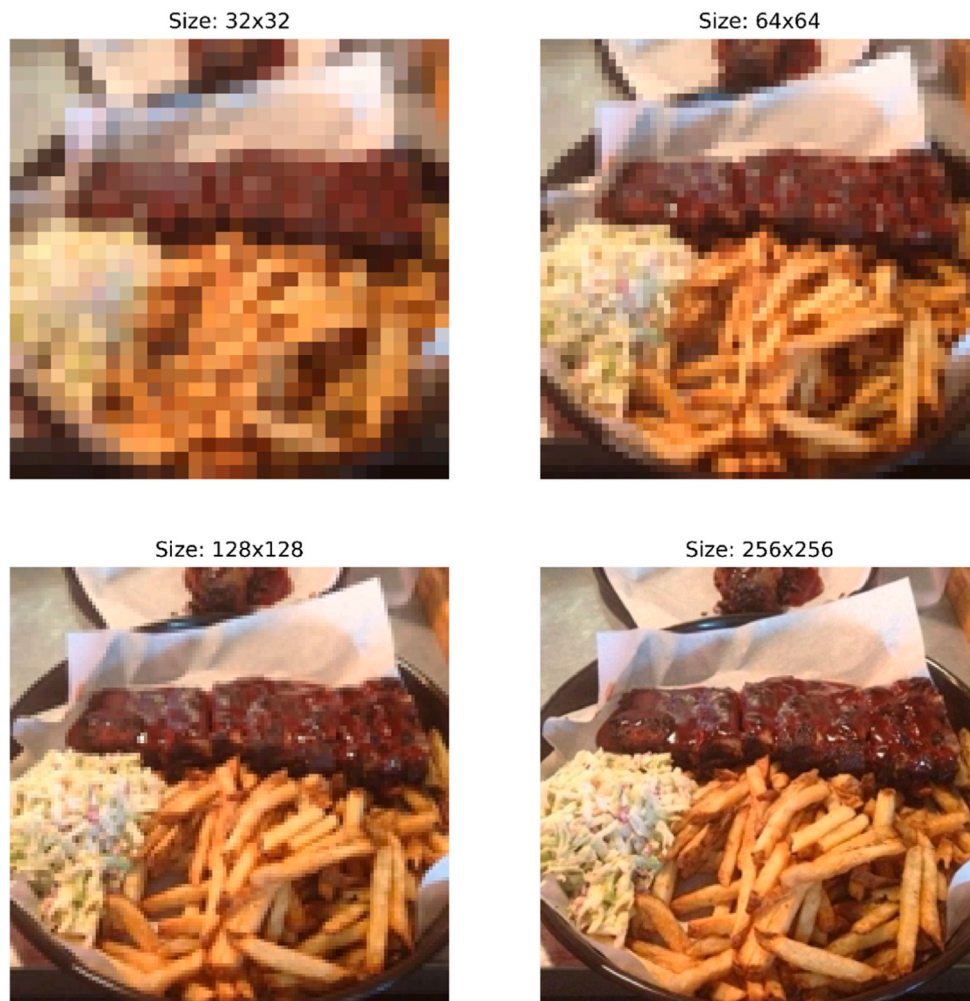


Fig. 2. Image representation across multiple resolutions.

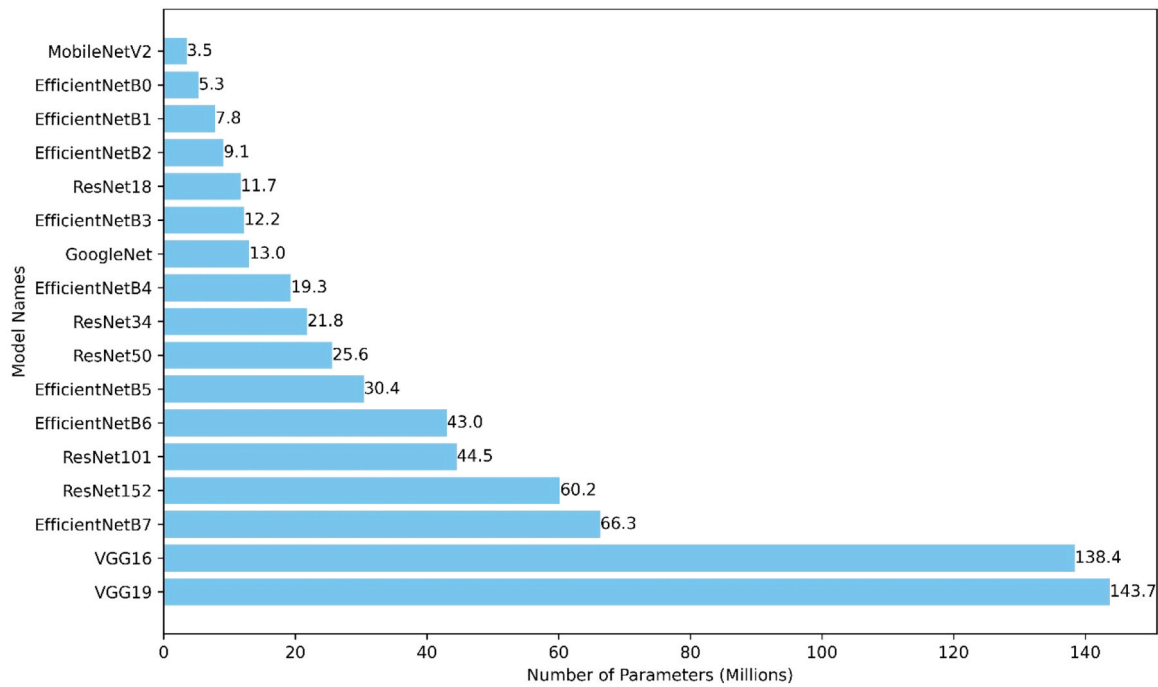


Fig. 3. Approximate parameter counts for some well-known models based on ImageNet.

2.7. Loss function, optimizer, and hyperparameter tuning

In this multiclass classification task, Cross Entropy Loss was used as the loss function and Stochastic Gradient Descent (SGD) as the optimizer. Several hyperparameters were tuned, including learning rate (LR), momentum, L2 regularization, and Nesterov accelerator for optimal results.

Momentum was set at 0.9 to help avoid local minima and achieve minimized loss and optimal performance. The Nesterov accelerator was enabled for better convergence. Learning rate, the most influential hyperparameter, was carefully chosen to ensure smooth convergence, avoiding divergence, and unnecessary fluctuations. Initially, LR= 0.001 was found to provide better and faster convergence. L2 regularization, also known as weight-decay in Pytorch, important for reducing overfitting and improving generalization, was also tuned. To reach optimal performance a grid of different learning rates and L2 regularization values was searched and LR= 0.001 and WD= 0.0001 were found to be suitable for all image resolutions in this study.

To even further optimize, a learning rate scheduler was used to dynamically adjust the learning rate. Through experiments, we found it beneficial to reduce the learning rate by a factor of ten every 10 epochs. The Training, Validation, and Evaluation batch sizes that were used were 64, 128, and 128 respectively.

2.8. Robust generalization & alternative for K-fold cross validation

Addressing real-world problems requires generalization and applicability. To ensure generalizable results with the Food11 dataset, which had been inherently pre-divided into training, validation, and testing subsets and did not permit K-fold cross-validation, a method inspired by 5-fold shuffled cross-validation was used. The code was executed five times from scratch for each image size and the average performance was reported. Each execution involved a fresh initialization of all parameters, the loss function, the model, and the optimizer, preventing data leakage. This approach improves the study's generalizability by addressing randomness in data transformation and data loader creation. It should be noted that the results, derived from executing the code five times, were anticipated to be similar for each image size.

2.9. Employed evaluation metrics

The utilized Food11 dataset was moderately, not extremely, imbalanced. We found accuracy sufficient for our needs, as other metrics like Precision, Recall, and F1 score are more relevant in cases of extreme imbalance or when certain misclassifications have severe consequences. However, a normalized confusion matrix was also used to identify common misclassifications and to provide comprehensive and understandable insights, which can be discussed further.

3. Results and analysis

Table 2 succinctly encapsulates the study's outcomes, emphasizing the evaluation subset's accuracy and speed across varying image dimensions. For brevity and clarity, we confined our report solely to the evaluation subset, representing unseen data that the model comes across in the real world. Each image size underwent five experiments and the mean values were computed. For deeper comprehension, Fig. 4 depicts the accuracy progression trend for various image sizes across 30 epochs, offering insights into model fitting, smooth convergence, optimal performance, employed techniques' impacts, and the influence of image

Table 2
Findings in terms of accuracy and speed for evaluation subset.

	Evaluation accuracies for 5 experiments (percent)	Evaluation speeds for 5 experiments (Image/s)	Average evaluation accuracy (percent)	Average evaluation speed (Image/s)
Image size =32	59.81 – 61.06 – 59.71 – 60.11 – 60.18	683.5 – 698.9 – 697.6 – 725.7 – 706.5	60.17 %	702.4
Image size =64	77.86 – 77.69 – 77.96 – 76.89 – 76.91	611.8 – 636.2 – 637.7 – 627.2 – 620.8	77.46 %	626.7
Image size =128	87.78 – 88.41 – 88.12 – 88.68 – 88.02	451.8 – 445.5 – 454.4 – 450.9 – 458.2	88.20 %	452.1
Image size =256	92.98 – 92.81 – 93.05 – 92.71 – 93.29	293.1 – 293.1 – 289.3 – 291.9 – 288.1	92.97 %	291.1

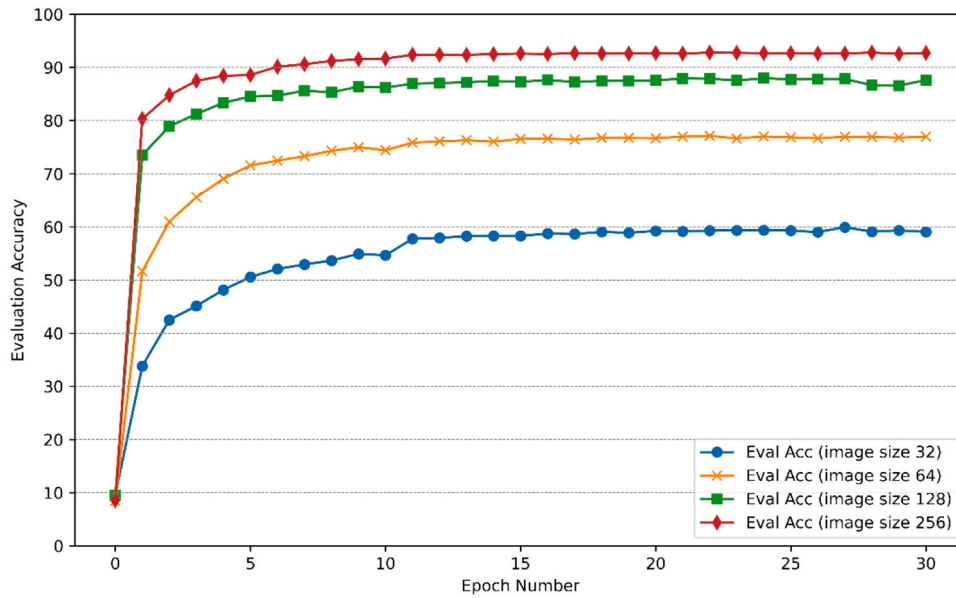


Fig. 4. Evaluation accuracy progression trend across 30 epochs for different image resolutions.

resolution.

Table 2 reveals that larger image sizes improved accuracy but decreased speed, increasing time. Notably, utilizing 256 * 256 images, the lightweight MobileNetV2 model could classify 291 images with ~92.97 % accuracy in a second, balancing speed and accuracy effectively. Additionally, both Table 2 and Fig. 4 show that accuracy gains, the amount of rise caused by using higher image sizes, decreased when image quality increased. Hence, a much smaller improvement in accuracy with 512 * 512 images compared to the ~5 % improvement from 128 * 128 to 256 * 256 images is expected. Table 2 also indicates that while for higher image resolutions accuracy gain between two consecutive image resolutions gets smaller, speed reduction gets noticeably greater. Thus, extremely high-quality images should be avoided for this application.

Fig. 5 presents the confusion matrix for the best image size of 256 * 256. It must be stated that this matrix was generated from a single code execution, and due to the inherent randomness, minor variations might occur in the matrices and internal accuracies. For example, some classes' accuracies may fluctuate up or down to a maximum of 2 %. However, these variations were found to be nearly negligible for evaluating and interpreting the results, specifically for higher image sizes. This ensures the validity and applicability of the results and subsequent discussions. Table 2 also confirms this point, showing that the disparity in average accuracy, the difference between minimum and maximum values for each image size, for image sizes 32, 64, 128, and 256 was 1.35 %, 1.07 %, 0.9 %, and 0.58 % respectively, indicating that higher image qualities also reduced the effect of randomness.

Fig. 5 reveals where misclassifications lay, providing more details. Initially, Fig. 5 illustrates that predictions were highly satisfactory, impressively 8 out of 11 classes surpassed 90 % accuracy and 10 out of 11 surpassed 85 % accuracy. Moreover, it offers that some classes, such as "Soup", "Rice", and "Noodles-Pasta", were significantly better predicted than some other classes like "Dairy-Products", "Bread", and "Dessert". It seems as if some classes were easier to predict while others were more challenging. Additionally, underperformed classes did not only show more misclassifications but also showed most misclassifications among themselves. For instance, "Dairy-Products" were more likely to be wrongly predicted as "Dessert". Since these points of contention can impact model performance, they should be discussed in depth.

4. Discussion

4.1. Studying the effectiveness of employed techniques

4.1.1. Transfer learning

The influence of transfer learning first became apparent during hyperparameter tuning. An initial learning rate of 0.001 was deemed suitable for the early epochs. This small value was a result of utilizing the pre-trained model. Without transfer learning, the anticipated learning rate would have been greater, possibly tenfold. The second impact is noticeable in Fig. 4. At Epoch=0, all weights were random, leading to an accuracy of around 9 % for all image qualities, as expected for an 11-class classification (100/11). However, after just one epoch of pre-trained model and full fine-tuning, the accuracies dramatically increased, hitting almost 80 % for images of 256 * 256 resolution. This demonstrates the efficacy of transfer learning in achieving high performance while eliminating the need for additional time and costly hardware.

4.1.2. Data augmentation and regularization

Data Augmentation and L2 Regularization were used to prevent overfitting, improving model robustness and generalization. This allowed extended training without early stopping concerns. Fig. 4 confirms their effectiveness as evaluation accuracies showed no overfitting, allowing the model to train until reaching the optimal point. Smooth convergence is another consequence depicted in Fig. 4.

4.1.3. Dynamic learning rate

The initial learning rate was found 0.001 for optimal performance. All accuracies saturated after 10 epochs (Fig. 4), showing the need for a dynamic LR. Therefore, LR was reduced tenfold after 10 epochs. Thus, epochs 11–20 and 21–30 had an LR of 1e-4 and 1e-5 respectively. This increased the previously-saturated accuracy for all sizes at epoch 11. However, the gain was much smaller between epochs 20 and 21 due to the nature of small LR at 1e-5. The dynamic LR also helped to maintain smooth convergence until reaching the optimal point.

4.1.4. Validation of generalization

As highlighted in Sections 2.8 and 3, we validated the results for each image size through five independent attempts, each time initializing parameters from scratch to prevent data leakage and bias. We

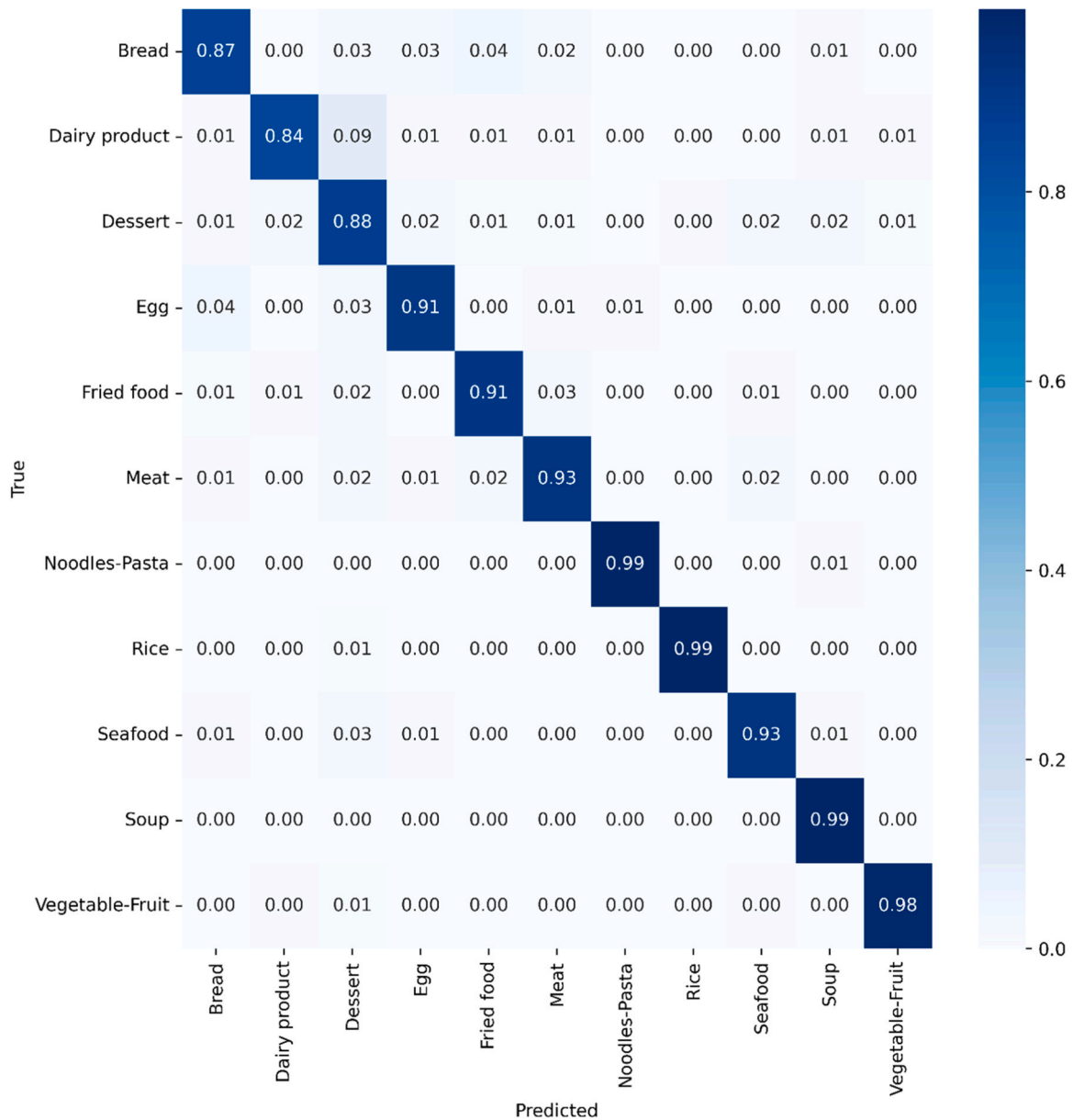


Fig. 5. Confusion matrix for 256 * 256 image resolution.

anticipated only minor variations due to randomness, and the consistent results for each image size, as illustrated in Table 2, confirm our findings' generalization in terms of accuracy and speed. Consequently, having generalized our results based on the available data, they are reliable and applicable for further research and practical applications.

4.2. Studying image quality

Table 2 shows that as image quality improved, accuracy rose but speed dropped. However, higher-quality images yielded less accuracy gain but significantly reduced speed, emphasizing the need for optimal image quality selection. In this study, 256 was found to be the ideal balance between high accuracy and brilliant speed.

Table 2 also depicts that the effect of randomness is diminished by employing good-quality images. This is because by utilizing good-quality images CNN can extract better and more meaningful features, resulting in better performance, also being less affected by randomness. This is more evident for MobileNetV2 with a less dense algorithm and simpler convolution.

Fig. 4 illustrates that transfer learning led to better performance and faster convergence for higher-quality images. This is probably because pre-trained MobileNetV2 has been trained on ImageNet's good-quality images, providing better weights for higher image qualities.

4.3. Studying confusion matrix and dataset challenges

The Food 11 dataset, utilized for this study, poses a significant challenge due to the intra-category diversity and inter-category similarities. For instance, the "Fried Food" and "Meat" categories overlap in images like Fig. 2, which contains both french fries and meat but is categorized as "Meat". A small salad presence could also be biased toward the "Vegetable-Fruit" category. Furthermore, the "Bread" category contains numerous pizzas and normal pieces of bread simultaneously. These examples are very frequent, representing complexities in the dataset, making prediction challenging.

As delineated in Section 3, Fig. 5 reveals that certain classes, such as "Soup" and "Vegetable-Fruits", "Rice", and "Noodles-Pasta" were classified accurately. Conversely, classes like "Bread", "Dairy Products", and

“Dessert” exhibited lower performance relatively. Notably, “Dairy Products” were frequently misclassified as “Desserts”, which adversely impacted the overall accuracy. Despite the common perception of neural networks as ‘black boxes’, insightful discussions and recommendations can indeed be beneficial.

Data imbalance and scarcity are critical factors to consider, significantly affecting the model’s learning process and class performance. More training data of a specific class generally improves the feature extraction and learning process, increasing the accuracy of that class. However, the observed misclassification pattern did not appear to be directly attributable to this phenomenon. Table 3 shows that “Noodles-Pasta” and “Rice”, despite their small contributions, performed excellently, matching “Soup”, which has the largest proportion. Moreover, although “Soup” and “Desserts” contributed evenly by 15 %, they performed completely differently. Also, “Noodles-Pasta” and “Dairy-Products” contributed almost evenly, but had the most and least performance respectively. There are also more examples to consider. Hence, at least in this study, prediction accuracies for a class do not heavily rely on the data volume phenomenon.

Our hypothesis attributes the observed differences to the food categories’ color patterns and overall visual characteristics. For example, the watery nature of “Soup” simplifies its classification. “Vegetables-Fruits” often exhibit green hues or specific colors common to most fruits. Similarly, the predominance of brown, black, and red in “Meat” facilitates its classification. “Rice” is grainy and comes mostly in white or brown colors, which facilitates its classification. Also, “Noodles-Pasta” can be assumed as a filamentary food category. Therefore, these categories tend to perform well.

Conversely, food categories such as “Bread”, “Dairy Products”, “Desserts”, “Egg”, and “Fried Food” typically exhibit bright color patches of yellow, orange, light brown, and white. These similar colors and patterns also exacerbate inter-category similarities. If our hypothesis holds, these classes are not only anticipated to yield lower performance but also are prone to mutual misclassification. This is particularly more relevant when employing MobileNetV2, which utilizes simpler convolutions compared to more complex models.

Table 3’s outcomes support our conjectures, but it is crucial to remember that these are hypotheses, backed by dataset understanding and observations. While the hypothesis aligns with the results, it may not be the sole determinant.

4.4. Comparison of results with similar studies

This section delves into research endeavors that parallel our objective of accurate Food-11 image classification. The investigation (Özsert Yigit & Özyildirim, 2018) utilized CaffeNet and AlexNet for food image classification. Using Stochastic Gradient Descent (SGD) as an optimizer, they attained accuracies of 80.51 % and 82.07 % respectively.

Table 3
The proportion of categories in the dataset and their corresponding accuracy.

Food Category / Classes	The approximate proportion in the dataset (also the approximate portion in each subset) (Table 1)	The approximate accuracy achieved on 256-sized images (Fig. 5)
Bread	10.4 %	87 %
Dairy	4.3 %	84 %
Products		
Dessert	15 %	88 %
Egg	9.9 %	91 %
Fried Food	8.8 %	91 %
Meat	13.2 %	93 %
Noodles-Pasta	4.4 %	99 %
Rice	2.8 %	99 %
Seafood	9 %	93 %
Soup	15 %	99 %
Vegetable-Fruit	7.1 %	98 %

Interestingly, when they switched to Adam as the optimizer, the same models demonstrated improved accuracies of 83.7 % and 86.92 % respectively. The image resolution used in their study was 512 * 512.

In a study (Islam et al., 2018b), the researchers utilized a pre-trained InceptionV3 model coupled with an effective data transformation technique for food image classification of the Food-11 dataset. The images used in this study had dimensions of 299 * 299 after pre-processing. The researchers were successful in achieving an accuracy of 92.86 %. In a subsequent research (Islam et al., 2018a), a CNN was built from scratch. Pretrained InceptionV3 was also used as a part of the study and the image dimensions in this study were 224 * 224. After applying various techniques, the proposed method yielded an accuracy of 74.7 %.

In this research (Suddul & Seguin, 2023), the initial approach was to construct a CNN from the ground up. However, this approach yielded a subpar accuracy of 64.6 %. To enhance their results, they incorporated a pre-trained EfficientNetB2 model, leveraging the benefits of transfer learning. Additionally, they employed data augmentation strategies to manage the imbalanced dataset and prevent early stopping. Ultimately, they achieved 94.5 % accuracy and 94.7 % F1 score.

In the recent research (Bu et al., 2024), combining data transformation, transfer learning, and ensemble learning models: VGG19, ResNet50, MobileNetV2, and AlexNet, all pre-trained on ImageNet, the researchers could reach a perfect accuracy of 96.88 %, surpassing all prior studies on the Food-11 dataset.

Leveraging a range of prestigious deep learning techniques and considering multiple image resolutions, our focus was on achieving harmony between performance and speed using lightweight pre-trained MobileNetV2. This approach enabled us to attain an impressive ~92.97 % accuracy for real-time analysis in practical applications using 256-sized images. Not only performance but also time and speed were key considerations in our approach. Approximately 88.2 % accuracy was also achieved using 128-sized images. We also investigated the effectiveness of employed techniques, remarkable misclassifications, and their probable reasons.

4.5. Policy implications

4.5.1. Possibilities for real-world applications

Given the impressive speed and high accuracy demonstrated, the study’s results could pave the way for the development of affordable devices for food recognition in food production and consumption scenarios. This has significant implications for tracking nutrition, food, and food waste. For example, it can be extremely advantageous for AI-operated kitchens that prioritize swift processes and minimal food waste, a feature sought after by many developed countries (Chakraborty & Aithal, 2024). As suggested (Gao et al., 2019; Zhu et al., 2020) cameras in smart refrigerators can play a pivotal role in food waste reduction and sustainability, and this research indicates that pre-trained MobileNetV2 is a viable option.

Additionally, the role of technology and mobile applications in the contemporary world is undeniable (Moumane & Idri, 2018), significantly facilitating and accelerating tasks across various sectors, including healthcare (Moumane & Idri, 2023) and food and diet management (Lubura et al., 2022). In this study, a pre-trained MobileNetV2 model with 256-sized images achieved approximately 93 % evaluation accuracy at very high speed. This performance and speed make the model suitable for many mobile applications and embedded devices, where access to expensive hardware is limited. The model requires training on a large dataset only once; thereafter, it operates in inference mode, enhancing speed by eliminating the backpropagation process, thereby emphasizing the importance of evaluation speed and performance.

4.5.2. Considerations

Although the results suggest several practical applications that can enhance productivity and efficiency, some aspects need to be addressed.

Firstly speed and time were obtained using a T4 GPU from Google Colab, so the speed is dependent on this factor. For instance, an A100 GPU, though more expensive, can provide significantly faster processing, while CPUs can significantly slow it down. Therefore, balancing speed and affordability is crucial. This consideration is especially important for mobile software or other embedded devices where low processing requirements and speed can enhance software performance (Moumane & Idri, 2017).

Furthermore, for real-world applicability, enhancing both model performance and efficiency can be achieved by using a more detailed dataset. For example, food categories like “Dessert,” “Fried Food,” and “Seafood” should be further subdivided into specific subtypes. Training a model on these detailed categories generally improves accuracy. Additionally, employing larger, more diverse, and detailed datasets is crucial, as it enhances performance and generalization.

Moreover, generalization and robustness should be improved by K-fold cross-validation or similar techniques like what we used in this study. This is crucial for real-world applications. In our study, we performed multiple runs to minimize the effects of randomness from data augmentation and data loader creation, also reporting the mean value for enhanced generalization.

4.6. Suggestions for future work and further improvement

- More future research can concentrate on accurate and real-time food recognition, utilizing or developing lightweight models or even deep dense models like EfficientNets or ResNet101 with suitable transfer learning methods.
- Investigating misclassified classes due to factors like similar patterns, colors, and data scarcity is pivotal. Research should also focus on understanding and addressing inter-category similarities and intra-category diversities.
- Ensemble learning, known for its improvement, could be valuable. Using multiple lightweight models together may enhance performance without considerably increasing time.
- Applying knowledge distillation to lightweight models can improve performance without significantly slowing down the speed, making it a compelling research area in image classification.
- This research centered on automatic food recognition, a classification task. Other computer vision methods like detection and segmentation could be employed to tackle global issues like food waste.

5. Conclusion

This research leveraged a pre-trained MobileNetV2 model, complemented by full fine-tuning, data augmentation, hyperparameter optimization, dynamic learning rate, and the use of images of varying quality for automatic, accurate, and real-time food recognition. The adopted approach proved effective in mitigating overfitting, reducing performance variability, and enhancing the model’s overall performance and generalizability. The findings revealed that using images of size 256, the model could classify numerous images per second with an accuracy rate of approximately 92.97 %. According to the results, using images larger than this size will not significantly improve accuracy as accuracy gains diminish for very high image qualities, yet speed drops significantly. The study underscores the efficacy of cost-effective pre-trained MobileNetV2, coupled with appropriate techniques, in delivering high speed and accuracy. Despite the dataset’s inherent challenges, such as inter-category similarities and intra-category diversities, which were discussed in detail, the results are commendable and applicable to real-world, real-time applications.

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CRedit authorship contribution statement

Babak Teimourpour: Writing – review & editing, Validation, Supervision, Project administration, Methodology, Formal analysis. **Shayan Rokhva:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Amir Hossein Soltani:** Writing – review & editing, Visualization, Validation, Resources.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used “Microsoft Copilot” in order to enhance language clarity and readability. After using this service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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